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The politics of baselines: A political feasibility framework for energy and climate scenarios

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ABSTRACT

Energy, transport, and climate scenarios rely on assumptions about future taxes, subsidies, and regulations, but such policy trajectories are rarely screened for political feasibility before they enter models. This paper addresses this often-neglected step in scenario design by developing a political feasibility framework to help modellers define and compare contested policy trajectories before quantification. Building on the distinction between outside and inside views of feasibility, the framework proceeds in two steps. First, it scopes a limited set of plausible trajectories using historical precedent, institutional practice, and actor commitments. Second, it evaluates them across five feasibility dimensions informed by institutional theory and political economy. The framework is demonstrated using a Swedish rail-demand forecast from 1999, in which a flat real road fuel tax path was used in the adopted-policy baseline. The case illustrates how seemingly neutral baseline assumptions embed political judgements and how a structured feasibility assessment can make alternative policy trajectories explicit and comparable. A simplified demand model is used to simulate the effect of alternative tax trajectories on rail demand. The framework offers scenario teams and modellers a practical approach to selecting and documenting policy assumptions based on political feasibility across adopted-policy, planned-policy, and exploratory scenario settings.

1. Introduction

Techno-economic models rely on assumptions about how taxes, subsidies, and regulations will develop over time. Yet the policy trajectories represented by these assumptions remain undertheorised in the literature on modelling future energy and transport systems. The issue extends beyond academic research. International organisations and national agencies routinely embed contested policy assumptions in scenario baselines and pathways. These assumptions shape model outcomes and can have consequential downstream effects when the results inform infrastructure planning, emissions projections, policy analysis, and political discourse. Such trajectories may be contested because they bring climate mitigation into conflict with competing goals, including economic competitiveness [1], household affordability [2], social acceptance [3], and energy security [4]. The factors that shape whether such policies can be adopted and sustained are well documented in the political economy literature [1,5–7]. What is missing is a structured way to assess politically contested policy trajectories before they enter models, in a form that modelling teams can apply in real-world settings. This paper addresses this gap by developing a political feasibility

framework for scenario design.

The framework builds on recent work on the feasibility of climate mitigation. Jewell and Cherp [8] define a broad climate option as feasible when it is “do-able under realistic assumptions” and argue that a sound feasibility assessment should trace causal pathways, compare options across contexts and scales, and be reflexive about who has the capacity to act, i.e. the dimension of agency. Drawing on Kahneman and Lovallo, they distinguish between two perspectives for assessing feasibility: the outside view and the inside view. The outside view assesses proposals through historical or spatial analogies, whereas the inside view analyses the individual causal factors that make an option more or less do-able under realistic assumptions. Applied to whole mitigation pathways, however, the inside view quickly becomes unwieldy for scenario teams. The move made in this paper is therefore narrower and more practical: instead of assessing entire climate mitigation pathways, the paper limits its application of the inside/outside-view distinction to contested policy trajectories that enter models as parameters.

While the framework can be used to rank the feasibility of policy assumptions, its primary value lies in the process. The outside view forces the analyst or team to make explicit which policy trajectories are

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considered plausible and why, rather than allowing a single baseline interpretation to dominate out of convenience or precedent. The inside view then disciplines comparison through a common set of dimensions, making political judgement more structured and transparent. In team settings, this can help establish a shared basis of evidence for discussing contested assumptions, while also revealing where disagreement remains.

The framework is intended for ex ante use in scenario design. In this paper, however, it is demonstrated ex post on a Swedish passenger rail forecast from 1999: a documented forecasting miss in which static assumptions about road fuel taxes and rail subsidies became embedded in long-term planning [9,10]. The case is used to demonstrate how the framework can be applied in real-world modelling contexts routinely faced by both international bodies and national agencies.

The remainder of the paper proceeds as follows. The next section clarifies why policy assumptions are difficult to specify in scenario practice. It then reviews the feasibility literature, with particular attention to the outside/inside-view distinction and the role of actors, before presenting the political feasibility framework. The framework is then demonstrated on the historical Swedish rail case, followed by a discussion of its applicability, limitations, and implications for scenario practice.

2. Policy assumptions in scenario design

Setting policy parameters is a routine but consequential part of scenario design. Whether used by international bodies such as the International Energy Agency (IEA) and the Intergovernmental Panel on Climate Change (IPCC) or by national agencies, models depend not only on assumptions about technology and prices but on how taxes, subsidies, and regulations are specified over time. A prominent example is the IEA's World Energy Outlook, which is organised around scenarios such as *Current Policies*, *Stated Policies*, and the more exploratory *Net Zero by 2050* pathways, illustrating the extent to which modelled outcomes depend on the specification of assumed policy trajectories. Similarly, EU member states are required to construct energy and emissions projections under *With Existing Measures* and *With Additional Measures* frameworks, where measures refers to policies in a broad sense. Even in adopted- or planned-policy settings, this is not merely a matter of legal classification. Policy trajectories often remain open to interpretation as continuing, expiring, being revised, or being politically contested within the modelled time horizon.

A recent example of how this plays out is the Swedish Energy Agency's 2025 long-term energy projections, in which a full tax exemption for pure biofuels was assumed to remain in place throughout the adopted-policy scenario extending to 2060, despite its contingency on recurring EU approval. This parameter choice, implemented in the bottom-up energy system model TIMES-Nordic, resulted in substantially higher projected biofuel uptake in road transport by 2035 (Assumption 1 in Fig. 1) than if the exemption was not assumed to continue beyond the next approval point (Assumption 2). The choice produced materially lower projected GHG emissions and was contested by downstream modellers involved in national climate emissions reporting. The example illustrates that even in adopted-policy settings, the practical issue is not simply whether a measure is adopted in legal form, but how its trajectory should be treated over time.

Exploratory scenarios pose the same problem but in a different form. They often combine normative targets such as net-zero by 2050 with socio-political storylines as in the *Shared Socioeconomic Pathways* (SSPs), which link climate futures to assumptions about globalisation, inequality, and policy ambition [12]. But the policy assumptions embedded in those futures are not necessarily politically grounded. The storylines provide qualitative depth and internal consistency, but are not inherently required to build on feasible policy assumptions and have been critiqued for including mitigation pathways with limited political support [13].

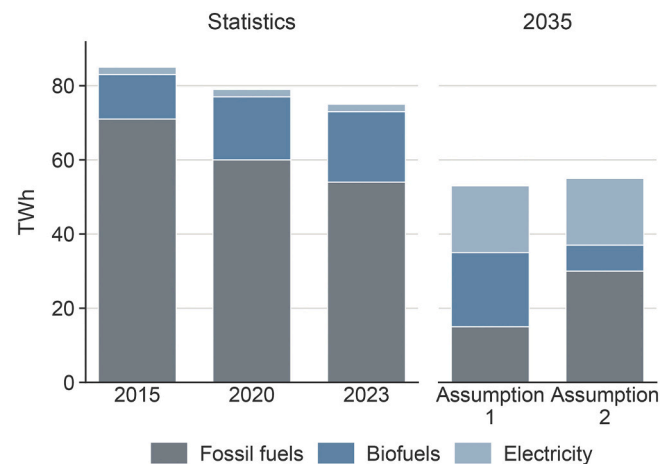


Fig. 1. Final energy demand from transport. Adapted from The Swedish Energy Agency [11].

As a result, there is a growing interest in integrating explicit political reasoning with quantitative models to bridge the gap between technically optimal pathways and what is practically and politically feasible [6,14,15]. From the perspective of scenario practitioners, the issue is not only that policy matters, but that policy trajectories often sit in an awkward space between legal definitions, political judgement, and model parameterisation. Without a structured way to specify them, modellers make implicit political judgements that are rarely transparent or reproducible. The feasibility literature offers conceptual resources for addressing this problem, but only partly solves the practical issue of operationalising feasibility in scenario design.

3. Current limitations

3.1. Feasibility and the representation of non-technical factors in models

Feasibility has been defined as the ability to implement a climate option under realistic assumptions [8]. Inderberg et al. [16] define political feasibility in actor-centric terms: an outcome or policy is politically feasible “if there is an agent or group of agents who have the capacity to carry out a set of actions which will lead to that outcome or policy in a given context”. Silvester [15] distinguishes political feasibility, dependent on human actions and interactions, from technical feasibility, which is dependent on other factors, and notes that political feasibility must consider actors, interest groups, and veto players. On a similar note, recent modelling scholarship has argued that greater realism in transition modelling requires better representation of actors, decision-making, and institutions, rather than relying exclusively on techno-economic factors [17].

Jewell and Cherp [8] also distinguish feasibility from probability and plausibility. Plausibility refers to the coherence of assumptions, while probability concerns the likelihood of outcomes. Feasibility, in contrast, refers to whether an outcome is do-able under realistic assumptions and by whom. This distinction matters here because many scenarios may be internally consistent in narrative or techno-economic terms while relying on policy assumptions that are institutionally fragile, politically contested, or weakly supported by relevant actors.

This point connects to a broader literature on the representation of non-technical factors in quantitative models. Recent reviews note that energy and climate models tend to emphasise techno-economic factors, while actors, institutions, behaviour, public acceptance, and political dynamics are excluded or represented through exogenous assumptions and qualitative discussion [17,18]. Krumm et al. [18], for example, show that social factors are often incorporated in storylines rather than as a structural part of modelling. De Cian et al. [17] similarly argue that model-based scenarios remain oriented toward normative outcomes,

with limited representation of the social interactions and institutional processes that lead to those outcomes. Socio-technical transition frameworks have been criticised on related grounds: by treating regimes as monolithic structures and the state as a passive recipient of policy direction, actor agency and political dynamics are rendered effectively exogenous to the analysis [19–21]. As noted by Jewell and Cherp [8], attempts to weave the political and the socio-technical into a “seamless web” risk oversimplifying political dynamics. The truncation of politics makes it difficult to specify who holds bargaining power, who will gain or lose from a transition, and how formal political rules shape feasibility. Political feasibility is therefore one important subset of the wider problem of how non-technical factors are represented in scenario design, and its inclusion requires analytical frameworks that keep actors and institutions at the centre.

3.2. The inside and outside views of feasibility

The distinction between outside and inside views in forecasting, originally proposed by Kahneman and Lovallo [22], is a useful step toward integrating feasibility assessment with modelling [8]. This distinction is also found in the megaprojects literature where it is used to formulate hypotheses about systemic cost overruns and delays in infrastructure projects [23]. The inside view treats climate mitigation or adaptation as a unique problem and seeks to identify the individual factors that influence mitigation pathways such as renewable energy deployment, including the role of land use, policy, and technological development. The outside view evaluates feasibility by examining reference cases and making comparisons, focusing on historical or spatial analogies available for the studied pathway.

The outside view is relatively straightforward to operationalise. Loftus et al. [13], for example, benchmark decarbonisation scenarios against historical rates of energy intensity improvement and technology deployment and find that the implied deployment rates outpace both historical cases and political support. Their contribution is not limited to political feasibility, but it demonstrates the outside view logic: feasibility can partly be assessed by comparing model assumptions and outputs with observed experience. Similarly, Goldstein and Qvist [24] argue for the role of nuclear power in climate mitigation by referencing rapid historical deployment rates in France and Sweden. While the outside view promotes comparability and anchoring of projections in real-world examples, it can be insensitive to differences in institutional and political contexts. The observed deployment rates for nuclear power in Sweden and France, for example, might not be transferable to other institutional and political contexts.

The greater difficulty, however, lies in operationalising the inside view. As noted by Jewell and Cherp [8], the approach is difficult to implement because it requires causal assessment across multiple domains and relies on expert judgement, which limits transparency and increases complexity. This becomes especially demanding when the object of assessment is not a discrete policy measure but a broad mitigation pathway involving interacting technologies, infrastructures, behaviours, institutions, and policies. When these are assessed simultaneously, the question “feasible for whom?” becomes clouded by complexity. Recent attempts to bring political feasibility closer to modelling illustrate both the promise and limits of the inside view. Inderberg et al. [16], for example, combine energy system modelling with analysis of actor support in order to identify politically significant model assumptions and locate key decision points in Norwegian transition pathways. Zhu et al. [14] similarly seek to improve policy realism by constructing politically feasible sectoral portfolios, implementing them in energy system modelling. These studies show that policy assumptions can be treated more explicitly and assessed endogenously within a modelling framework, while highlighting how actors shape outcomes. At the same time, they remain relatively demanding exercises, oriented toward broad, normative pathway assessment and country-specific policy and technology portfolios rather than offering a

generalised approach to setting policy parameters in conventional scenario practice. This limitation is echoed in the integration literature: studies linking socio-technical frameworks with quantitative models have similarly focused on normative decarbonisation goals and multi-dimensional transition pathways rather than on the feasibility of specific policies [25,26]. The result is that the integration literature has advanced the understanding of how transition processes can be modelled, but without producing a transparent and transition-agnostic workflow for assessing individual policy trajectories.

The review therefore points to two limitations that motivate the framework developed here. First, the outside and inside views provide complementary lenses, but the inside view becomes cumbersome when applied to whole mitigation pathways composed of many interacting variables. Second, the literature on integrating social and political dynamics into modelling has not mapped onto the needs of conventional scenario practice, remaining focused on broad mitigation pathways rather than the specific policy parameters that scenario practitioners deal with.

4. The framework: Method and application

4.1. Overview of approach

The framework has two linked functions. First, it helps define a limited set of plausible policy trajectories that are meaningful to compare within a given scenario setting. Second, it provides a common structure for comparing those trajectories through explicit political reasoning. The outside view performs the first task by scoping trajectories through precedent, institutional practice, and actor positioning. The inside view performs the second by comparing them across five dimensions of political feasibility. The division of labour is such that the outside view keeps scoping open enough to avoid tunnel vision, while the inside view disciplines judgement once comparison begins. Fig. 2 summarises this two-step workflow.

4.2. Step 1: Scoping candidate trajectories

The outside view is used to scope a limited set of candidate trajectories for later comparison. It draws on historical analogies, prior policy adjustments, announced reforms, institutional decision cycles, political discourse, and stakeholder signals. The aim at this stage is not to determine which trajectory is most feasible, but to define a manageable set of plausible trajectories expressed in modelling terms. In practice this step can be conducted by a single modeller or by a small team through document review, workshops, horizon scanning, expert discussion, and related scoping exercises. The data, materials, and insights gathered in this step can later be used as evidence in the inside view assessment.

The implications of the outside view scoping depend on the scenario framework. In an exploratory scenario, the candidate trajectories may simply define alternative policy futures to be screened in the inside view. In an adopted- or planned-policy scenario, however, the question is narrower: whether the baseline interpretation of adopted or planned policy is robust, or whether a more dynamic trajectory or alternative paths, based on precedent, policy direction, and recurrent decision practice, should also be treated as credible or at least justified as a sensitivity case. For example, if the contested policy parameter under consideration is a tax exemption for a fuel, the outside view might identify three trajectories: continued exemption, limited continuation and phase-out, or expiry at the next decision point. These trajectories might be derived from prior renewals, public commitments, comparable cases, and known institutional constraints. The practical value of the outside view in this case is therefore that it helps modellers move beyond narrow legalistic interpretations of adopted policy. For exploratory scenarios, it helps ground contested assumptions in structured political reasoning rather than storytelling.

For this step, it is useful to clarify the timelines that are applicable.

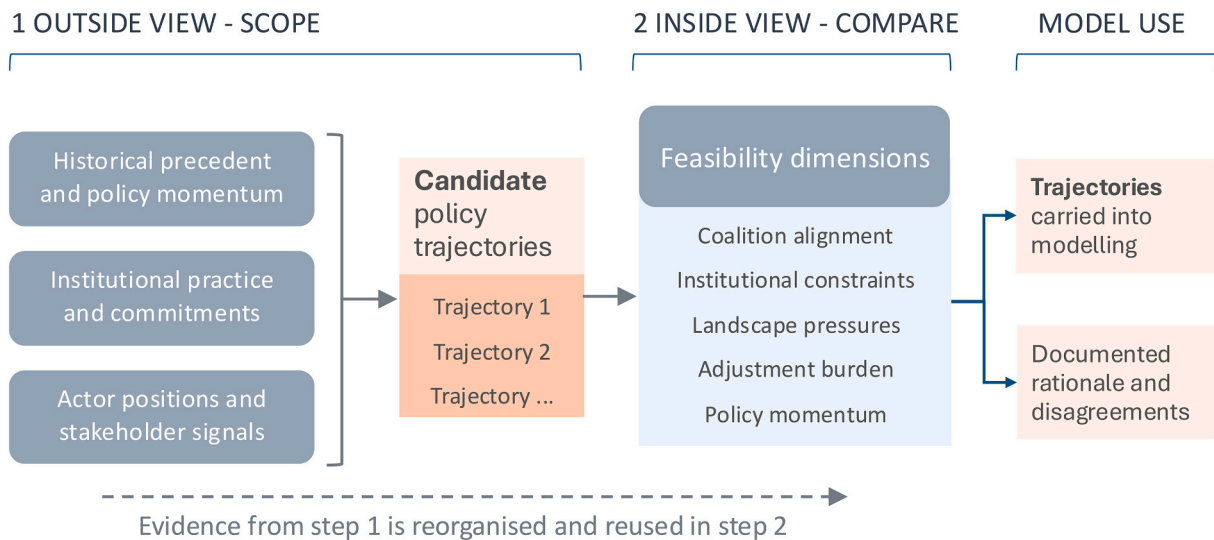


Fig. 2. Overview of the political feasibility framework and its two-step workflow.

The framework is designed for timelines where at least one consequential decision window falls within the modelling horizon. This is especially clear for the tax exemption discussed in Section 2, which is contingent on scheduled EU approvals. But decision windows may also be more diffuse. Consider a policy central to the EU's climate agenda: the adopted 2035 zero-emission target for new cars and vans. Although formally adopted, it remains politically contested and open to challenge and revision through several institutional and political channels leading up to its implementation. In such cases, the most consequential decision window can be identified and used to anchor the outside view scoping. The available evidence should also constrain this choice. The further the horizon extends beyond such a revisable decision point, the weaker the outside view becomes as a guide to feasibility. Over longer periods, the political economy of a policy and actor interests may shift [1], making historically grounded inference less secure.

A practical part of the outside-view scoping is to map the actors positioned around the trajectories under assessment. Consistent with an actor-centric framing of political feasibility, the question is who has the capacity and interest to support, block, or adapt to a trajectory, and how much leverage they hold [15,16]. The framework uses three categories as an organising shorthand grounded in incentive logic rather than formal transition theory. **Rule-keepers** are actors whose material or political interests align with preserving the current policy arrangements: typically incumbents, established industry associations, or bodies that benefit from existing arrangements. **Rule-switchers** are actors who stand to gain from change, such as advocacy coalitions, political parties, or firms that can exploit decision windows to advance policy change. **Rule-takers** neither drive nor resist change but signal how readily a new trajectory will be absorbed: their adjustment costs affect the political sustainability of reform [5,7]. These are not permanent classifications: the same actor may shift position across decision windows, and they are not designed to be exhaustive. Their function is to structure the initial actor survey, direct attention to where veto leverage and coalition support are concentrated, and carry that structured view into the inside view comparison.

4.3. Step 2: Assessment of political feasibility

The inside view forms the comparative core of the framework. It assesses each trajectory across five dimensions that capture channels through which a policy trajectory may be easier or harder to adopt, renew, revise, or sustain. **Coalition alignment** addresses how organised actors are positioned in relation to the trajectory and how much influence they wield. **Institutional constraints** capture formal rules, veto

points, and legal or fiscal conditions that enable or block the trajectory. **Landscape pressures** capture dominant political, economic, or cultural paradigms that can strengthen or weaken trajectories. **Adjustment burden** addresses whether visible losers can adjust or be compensated, in relation to their political leverage. **Policy momentum** captures whether the trajectory extends or departs from established policy direction, routines, and expectations.

The five dimensions (Table 1) are selected because they capture important channels through which policy trajectories are enabled or blocked, drawing on a body of institutional theory, socio-technical theory, and political economy literature. Coalition alignment draws on veto player and coalition theory: whether a winning coalition can be

Table 1
The five political dimensions of the political feasibility framework.

Dimension	Motivating question	Examples
Coalition alignment	What rule-keepers and rule-switchers are positioned around the trajectory, what do they win or lose materially or politically, and how much influence do they wield?	Government and support-party positions; budget agreements; mobilisation of business associations, unions, NGOs, affected voter blocs, and industries.
Institutional constraints	What formal rules, veto points, and legal or fiscal constraints enable or block the policy trajectory?	EU state aid rules; constitutional or budget-law limits; power of courts and regulatory agencies.
Landscape pressures	What influential macro trends push the trajectory in a certain direction?	International agreements and targets; energy crises or wars; shifts in dominant economic ideas; rising environmental concern in public opinion.
Adjustment burden	Are there credible mechanisms for losers and/or rule-takers to absorb change or be compensated for losses?	Availability of technical alternatives; transitional support and retraining schemes; targeted tax rebates; redistributive measures that offset visible losses.
Policy momentum	What existing policymaking practices, precedents, norms, and expectations reinforce or obstruct the policy?	Previous tax changes; long-standing subsidies or exemptions; sunk investments and business models from policy support; entrenched administrative practice.

assembled and whether actors with veto capacity oppose the trajectory are among the most reliable predictors of policy change [15,27,28]. Institutional constraints reflect the role of formal rules, budget procedures, and regulatory frameworks in conditioning what governments can and cannot do at a given moment [29,30]. Landscape pressures capture how macro-level trends, including international commitments, shifts in dominant economic and cultural ideas, and public opinion, shape policy trajectories [31]. Adjustment burden addresses the distributional dimension of reform: politically visible losses concentrate opposition, and feasibility improves when losers can adjust or be compensated [5,7]. Policy momentum captures path-dependent dynamics: a trajectory that extends established practice faces less resistance and carries more institutional credibility than one that represents a sharp departure [32]. Taken together, the five dimensions do not eliminate judgement, but they make reasoning that modelling teams often apply implicitly more explicit, comparable, and transparent.

4.4. Evidence gathering and synthesising

For each trajectory, the modeller or team gathers evidence from the types of material typically available at the time of scenario design: government bills, budget documents, agency reports, stakeholder consultation material, parliamentary debates, public statements, stakeholder documents, contemporary media coverage, and relevant secondary analyses. The same material gathered in the outside view can be reused here. The difference is that it is now organised according to the five political dimensions and used for comparative assessment.

To illustrate the coding logic using an example from this paper, a 1997 statement from the Swedish Employers' Confederation (SAF) opposing further carbon tax increases on the grounds of industrial competitiveness was coded as demonstrating rule-keeper behaviour under coalition alignment, indicating an actor with organised voice but without direct control over the budget agenda. This excerpt, together with similar statements from the motorists' lobby, established a pattern of organised opposition to further road fuel tax increases. The full coding table and actor mapping with documentary sources for the case study are provided in Appendix A.

4.5. Comparing trajectories

Depending on the purpose, composition of the modelling team, and available sources, the inside-view assessment may range from a structured qualitative appraisal by a single modeller to facilitated workshops, expert elicitation, ranking procedures, or, as in the present case study, pairwise comparison. Pairwise comparison is one option when several trajectories are assessed by one or more analysts. Comparing two trajectories at a time may reduce cognitive burden and make the evidence basis for judgement more transparent [33]. It can also reveal dimensions on which assessors disagree and form the basis for discussion.

4.6. Data sources

The framework is designed to use documentary material typically available at the time of modelling. In the case study, the documentary corpus consisted of government bills and energy propositions, parliamentary motions, annual budget documents, committee reports, agency publications, polling data, and media coverage. The corpus was restricted to documents in the public domain prior to the modelled decision window, covering the period from the 1988 transport reform through 1999. The outside view drew primarily on historical tax data and political documents outlining policy direction. The inside view reorganised the same material under the five political dimensions to support comparative assessment. A representative selection of coded excerpts and the resulting actor mapping are documented in Appendix A.

5. Case study: Swedish rail forecast

5.1. Motivation

This case study uses a Swedish rail forecast produced by the Samplan group in 1999 to demonstrate how the framework can be applied to a politically contested policy parameter. The focus is on the road fuel tax trajectory, including energy tax, carbon tax, and VAT, as a driver of substitution between road and rail transport in the Samplan model. The framework is designed to be applied *ex ante*, before modelling. However, for demonstration purposes, a documented historical forecast miss that relied on static policy assumptions is used as a case study, to show how the framework can be integrated into real-world scenario practice.

The 1999 forecast was used to evaluate alternative transport-policy strategies for long-term planning. The reference scenario, representing adopted policies, became the baseline for subsequent infrastructure planning and CBA-style assessments. The models optimised travel demand across modes including rail, road, and air. Although the travel modes were not perfect substitutes, changes in the cost of one travel mode affected demand for the others. Consequently, parameters such as fuel prices and taxes strongly influenced modal shares. The Samplan group projected a 31% increase in rail demand between the base year 1997 and 2010, whereas observed demand increased by 59% over the period. Later evaluations highlight two assumptions as particularly important for the forecasting miss: fixed fuel taxes and omission of expansion in regional rail services and subsidies [9,10]. The present application focuses on the first of these, since the road fuel tax trajectory is more easily parameterised in a demonstration of the framework.

Overlooking these two policy-driven dynamics led the forecasts to underestimate the pace of rail demand growth. The full extent of this growth is captured in Fig. 3: between 1990 and 2010, regional passenger rail increased from 2 to 5 billion passenger-kilometres, and long-distance rail from 4.6 to 6.1 billion [34].

5.2. Outside view scoping of candidate trajectories

This section demonstrates the outside-view component of the framework. This step was conducted by the author but could also be organised as a small team exercise through workshops and discussions. Its purpose was to identify a small number of candidate policy trajectories that were sufficiently grounded in historical development, public commitments, political debate, and the institutional context of environmental tax policy. In this case, the outside view drew on prior

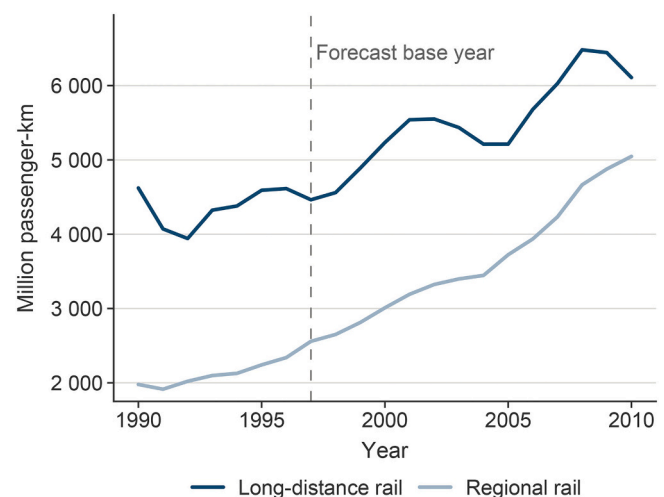


Fig. 3. Development of passenger kilometres on long-distance and regional rail in Sweden, 1990–2010. The Samplan model baseline is indicated for 1997. Source: Transport Analysis [34].

changes in Swedish fuel taxation, the policy direction expressed in government and parliamentary documents, and the annual budget cycle as the operative decision window.

The outside view scoping begins with the transport bill of 1988, which repositioned Swedish transport policy around socio-economic efficiency, environmental sustainability, and the explicit goal to internalise negative externalities from transport [35]. Infrastructure investments were to be made on socio-economic grounds, using cost-benefit analysis to identify projects that were welfare-maximising for society [36,37]. The broader direction of reform thus exposed road transport more clearly to arguments about external costs, while rail increasingly came to be framed as environmentally preferable.

The national policy direction was reinforced by wider developments during the 1990s. The Kyoto Protocol was agreed in 1997 and Sweden's own policy documents increasingly tied domestic environmental taxation to a wider international and European policy focus [38,39]. Proposition 1997/98:56 made explicit that transport policy should contribute to socially, economically, and ecologically sustainable development and restated that taxation in transport should reflect the polluter-pays principle. Proposition 1996/97:84 on sustainable energy supply similarly framed energy and carbon taxation as instruments for fuel switching and ecological sustainability, while also acknowledging the need to protect industrial competitiveness. In other words, the relevant policy environment was not one of stability, but of continued pressure toward environmental cost internalisation.

The political debate at the time also pointed in a more ambitious direction. The Social Democratic government was pursuing fuel tax increases as part of a long-term fiscal and policy strategy rather than as a one-off adjustment [40]. The Green Party pushed more explicitly for annual increases and a long-run green tax shift, making fuel taxes part of its bargaining strategy [41]. The Left Party was broadly supportive of stronger environmental taxation when combined with redistribution. As such, there was a broad governing coalition of actors in support of tax increases. Opposition was organised around the Swedish Employers' Confederation (SAF), parts of heavy industry, the motorists' lobby, and the Moderate Party, all with a stake in resisting further tax increases. But they did not amount to an effective veto or parliamentary coalition. The voters who stood to lose from tax increases were primarily motorists. Contemporary polling, however, showed broad public support in Sweden for environmental taxation [42].

The actor roles in Table 2 are assigned based on actors' revealed positions on the road fuel tax trajectory within the operative decision window, the annual budget process, rather than by actor type. The Social Democrats, the Greens, and the Left Party all favoured further increases and are therefore coded as rule-switchers relative to a continuation of the 1999 tax level. Parties that opposed further increases, most prominently the Moderate Party, took the rule-keeper role alongside business and motorists' organisations. Political parties differ from the other actors in the table in the nature of their involvement: they hold formal agenda-setting and voting power in the budget process, whereas organised interests exert influence through media and mobilisation, and by making visible cost increases politically salient.

The material collected during the outside-view scoping was reviewed to identify the range of politically plausible directions under discussion at the time. The actor mapping showed a clear rule-switcher coalition consisting of the Social Democrats, Greens, and Left Party with agenda-setting power over the annual budget, while rule-keepers in industry and the motorists' lobby had voice but lacked parliamentary veto leverage. Contemporary polling showed broad public support for environmental taxation. This suggests that, although motorists carried the direct adjustment burden, incremental increases did not appear to carry a significant political cost from public opposition.

This configuration helped inform the three candidate trajectories for subsequent comparison. Table 3 shows how T1 represents a strict legal definition of adopted policy: maintaining road fuel taxation at its 1999 level in real terms. T2 represents a continued incremental adjustment

based on policy momentum: the average increase between 1991 and 1999 was assumed to continue. T3 represents a stronger green-tax-shift path based on proposals advanced by actors who made government support conditional on environmental policy. The decision window is the same for all trajectories but could vary depending on the context. The parameterisation of the policy trajectories was based on the average annual increase between 1991 and 1999 for T2, while T3 was parameterised as an annual real increase of SEK 0.10 per litre, expressed in constant 1999 prices, drawing on a concrete proposal from the time [40].

5.3. Preparation of inside view material

For the case study, the evidence collected in the outside view was synthesised into pairwise comparison sheets. Each sheet compared two candidate trajectories at a time and summarised the evidence for each political dimension in a standardised format. The purpose was to condense the relevant evidence into a form that was transparent, comparable, and manageable for assessors. A short description of each dimension was drawn from Table 1. For each trajectory, the comparison sheets recorded the trajectory definition, a short evidence summary under each dimension, and an overall judgement. This gave assessors a compact and transparent basis for judgement while keeping comparison manageable. Table 4 provides the format and content used for the comparison between a flat real road fuel tax (T1) and a moderate annual increase (T2) for the case study.

In the case study, three trajectories were identified requiring a total of three pairwise comparisons: T1 versus T2, T1 versus T3, and T2 versus T3. For each pair, the assessor is first asked to compare the two trajectories dimension by dimension, indicating which appears more feasible on that dimension or whether no clear difference can be identified. The assessor then records an overall comparative judgement for the pair and highlights the most influential dimensions. In practice, this can be done by an individual analyst or by a small team as demonstrated in the case study. In team settings, individual judgements can be compared and discussed before recording a final comparative assessment.

5.4. Inside view assessment: Pairwise comparison

To demonstrate the framework using the case study, the inside view assessment was conducted by four researchers involved in energy modelling, only one of whom was an energy-policy expert. This illustrates one possible mode of application. In other settings, the assessment can be conducted by a modelling or research team, or by a single modeller, using alternative assessment methods that vary in scope and complexity. The group was introduced to the modelling setup and political context of the era, as if assessing the case in 1999. They were introduced to the three trajectories identified in the outside view (Fig. 4), and the motivation for their inclusion. Each assessor received the same three pairwise sheets together with the actor sheet shown in Table 2. The researchers were introduced to the political dimensions and the logic behind them and were allowed to ask questions about their interpretation. Following this introduction, each person was asked to work through the sheets individually to choose the most feasible trajectory on each dimension, and to make an overall comparative judgement of each pair. They were asked to highlight the dimensions that were most influential in their choice. After completing the assessment, the group held a discussion to explore agreements and disagreements.

As shown in Table 5, the pairwise comparisons pointed to T2 as the most feasible overall trajectory, with pairwise wins over both T1 and T3. The identification of T2 as the most feasible was therefore consistent across assessors. The T1-T3 comparison was evenly split across assessors. The assessors cited the trade-off embedded in the choice: T1 imposed a lower adjustment burden on motorists, whereas T3 was more closely aligned with policy momentum and broader landscape pressures. The assessment therefore produced a consistent preferred trajectory

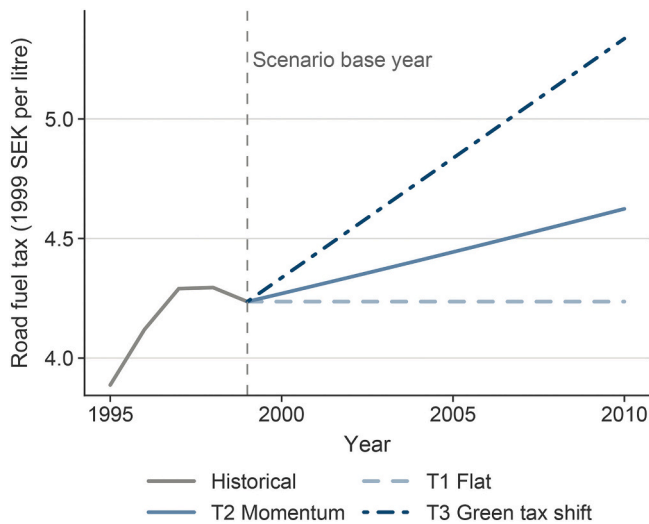


Fig. 4. Candidate road fuel tax trajectories. Swedish kronor per litre of fuel in constant 1999 prices. Actual road fuel tax data adapted from The Swedish Environmental Protection Agency and The Swedish Energy Agency [43].

within this exercise, T2, but not a complete and uncontested ordering of all trajectories. Even if the pairwise assessment had produced a perfect comparative ordering of the three candidate trajectories, this ordering would not by itself determine the baseline assumption. In the present adopted-policy setting, the comparison is used to assess whether the flat road fuel tax (T1) represented a defensible interpretation of adopted policy, or whether the definition should have been expanded to better accommodate the intended use of the scenario, which included infrastructure planning.

5.5. Simulation of policy trajectories

To illustrate the modelling implications of the feasibility assessment, the three road-fuel-tax trajectories are simulated over 1999–2010. The stylised demand model isolates the road-fuel-cost effect by holding the

Table 3

Candidate trajectories identified in the outside view scoping.

Trajectory	Definition	Basis	Decision window(s)
T1	Flat real road fuel tax	Mirrors Samplan baseline and adopted-policy interpretation at the model cut-off date	Annual national budget cycles
T2	Moderate annual real increase	Continues recent historical tax adjustment and aligns with government policy direction	Annual national budget cycles
T3	Stronger green-tax-shift path	Reflects proposals and discourse for a more ambitious green-tax shift	Annual national budget cycles

real non-tax fuel price, VAT rate, demographic development, income, regional integration, and rail-service supply constant. The published Samplan forecast uses 1997 as the model base year, while 1999 is used here as the common starting point from which the counterfactual tax trajectories diverge. This separates the model's calibrated base year from the point at which the policy assumptions are specified, consistent with routine scenario practice. The tax is a weighted average of petrol and diesel taxation, using weights of 0.9 and 0.1, respectively, to approximate the fuel composition of passenger-car travel.

Let $\tau_{s,t}$ denote the weighted real road fuel tax per litre of fuel under trajectory s , where $s \in \{T1, T2, T3\}$, and let $\tau_0 = 4.236$ denote its value in 1999, expressed in constant 1999 SEK:

$$\tau_{s,t} = \begin{cases} \tau_0 & s = T1 \\ \tau_0 (1.008)^{t-1999} & s = T2 \\ \tau_0 + 0.10(t - 1999) & s = T3 \end{cases}$$

The corresponding consumer fuel price is given by:

$$P_{s,t} = (c + \tau_{s,t})(1 + v)$$

where $c = 3.50$ SEK per litre is the real non-tax fuel price and $v = 0.25$ is the VAT rate. VAT is applied to both the product component and the fuel

Table 2

Actor overview for the road fuel tax policy trajectory.

Actor	Position on road fuel tax trajectory	Role	Why this actor matters
Social Democratic government	Favours continued fuel tax increases for fiscal reasons.	Rule-switcher	Controls the budget agenda and can advance tax changes through ordinary budget processes.
Green Party	Favours a stronger green tax shift and recurring fuel tax increases.	Rule-switcher	Pushes for a more ambitious environmental tax trajectory.
Left Party	Generally favourable if combined with redistribution or social balancing.	Rule-switcher	Supports reform, but distributional concerns matter.
Moderate Party	Opposes further fuel tax increases.	Rule-keeper	Parliamentary opposition with formal voting power, but without a blocking majority within the immediate decision window.
SAF / business interests	Opposes further fuel tax increases.	Rule-keeper	Represents organised opposition based on competitiveness concerns.
Motorists' organisations	Oppose further fuel tax increases.	Rule-keeper	Make visible cost increases politically salient.
Households / motorists	Mixed; exposed to cost increases but with some support for environmental taxation.	Rule-taker	Their acceptance affects how politically costly increases become.
Environmental NGOs	Support continued and stronger environmental taxation.	Rule-switcher	Reinforce the legitimacy of tax increases in public debate.

Table 4

The assessment overview used in the inside view pairwise assessment of political feasibility.

Field	T1: Flat real road fuel tax	T2: Moderate annual real increase	Comparative judgement
Trajectory description	Real road fuel tax remains constant in real terms from the 1999 level.	Real road fuel tax continues to increase gradually in real terms, broadly in line with the recent historical trend.	
Coalition alignment: Which trajectory is backed by the more influential coalition of actors within the decision window(s)?	Favoured by actors seeking to prevent further tax increases, including motorists' organisations, business interests, and tax-sceptical opposition actors. These actors have public voice, but limited direct control over the budget agenda.	Favoured by the governing and support-party constellation around environmental taxation, especially actors already committed to incremental tax increases and ecological tax reform. This coalition has agenda-setting and bargaining leverage in the budget process.	T1 / T2 / No difference
Institutional constraints: Which trajectory faces fewer formal or procedural obstacles?	Can be maintained through ordinary annual budget decisions.	Can be maintained through ordinary annual budget decisions.	T1 / T2 / No difference
Landscape pressures: Which trajectory is more consistent with broader societal and political trends, including public opinion, discourse, and external commitments?	Sits within a context of growing climate and environmental concern, but does not extend that direction through further tax increases.	Sits within a context of growing climate and environmental concern and continued use of environmental taxation. Polling indicates broad support for environmental taxation.	T1 / T2 / No difference
Adjustment burden: Which trajectory creates fewer losses that are challenging to absorb or adjust to?	Avoids additional visible costs for motorists and other fuel users, reducing the immediate need for compensation or adjustment measures.	Imposes gradual additional costs on motorists and other fuel users.	T1 / T2 / No difference
Policy momentum: Which trajectory better continues recent policy practice and expectations?	Represents a break from the recent pattern of fuel tax adjustments during the 1990s.	Continues the incremental adjustment pattern visible during the 1990s and adheres to broader policy commitments.	T1 / T2 / No difference
Overall comparative judgement			T1 / T2 / No difference

tax. The real non-tax fuel price and VAT rate are held constant across trajectories.

All three trajectories have the same fuel price in 1999, while the price under T1 remains constant through 2010, mirroring the Samplan assumption. The price-induced rail-demand multiplier for each trajectory relative to T1 at the forecast endpoint is therefore:

$$M_s = \left(\frac{P_{s,2010}}{P_{T1,2010}} \right)^{\varepsilon_{\text{cross}}}$$

where $\varepsilon_{\text{cross}} = 0.35$ is the cross-price elasticity of rail demand with respect to road fuel prices. The elasticity is not stated explicitly in the Samplan documentation, but the value used here is consistent with recent estimates and reviews [44,45]. Expressing the multiplier relative to T1 makes it possible to represent the incremental effect of the alternative policy trajectory rather than total rail-demand growth over this period.

The published Samplan forecast reports a 31% increase in rail-passenger demand between 1997 and 2010 [37]. Normalising rail demand in 1997 to one therefore gives a baseline demand index of 1.31 in 2010. Applying the price-induced demand multiplier to this endpoint, the adjusted growth under trajectory s is:

$$\hat{g}_{s,1997-2010} = 1.31M_s - 1$$

For T1, $M_{T1} = 1$, so the equation reproduces the original 31% forecast.

The central elasticity represents a long-term response to road fuel costs. A sensitivity interval of 0.15–0.45 spans lower public-transport estimates and higher longer-run estimates from the rail-demand literature. The weighted pump price $P_{s,t}$ was reconstructed with historical data on fuel taxation [43].

As shown in Table 6, the simulations show that the alternative road fuel tax assumptions affect the projected increase in rail demand, while a

substantial part of the difference between forecasted and observed growth remains associated with other assumptions. T2 increases projected rail-passenger growth from 31.0% to 33.3%, an uplift of 2.26 percentage points relative to T1. This uplift corresponds to 8.1% of the observed forecast gap. For T3, the corresponding share is 22.3%.

Across the sensitivity interval of 0.15–0.45, projected rail-passenger growth ranges from 32.0 to 33.9% under T2 and from 33.6 to 39.1% under T3. The remaining difference between these simulations and observed growth points toward the role of additional developments over the period, including population and income growth, regional integration, rail subsidies, service frequency, and network expansion, several of which were identified as problematic policy assumptions in subsequent reports [9,10], suggesting those assumptions could have benefited from a similar feasibility treatment.

6. Discussion

The case study draws attention to the distinction between the legal status of a policy assumption and its wider political context over the modelling horizon. Extending the flat road fuel tax throughout the period implied that it would remain unchanged through repeated budget decisions, despite a history of gradual increases, environmental commitments, and an organised coalition of actors favouring further increases. The question is therefore whether a strict legal interpretation at the starting point provides a neutral baseline for an adopted-policy scenario, particularly when model outputs feed into investment decisions, infrastructure planning, and downstream energy-system modelling or emissions projections.

The distinction between legal status and feasibility also helps motivate the contribution of the framework in relation to the broader feasibility literature. The inside and outside views have primarily been used to consider whether mitigation options or transition pathways are do-able under realistic conditions. At the level of a complete pathway,

Table 5

Results from the inside view pairwise comparisons conducted for the case study.

Pair	Assessor 1	Assessor 2	Assessor 3	Assessor 4	Comment
T1 vs T2	T2	T2	T2	T2	T2 was seen as having broad coalition support and representing a feasible continuation of current momentum while T1 lacked alignment with landscape trends and momentum.
T1 vs T3	T1	T3	T1	T3	T1 imposes lower costs on motorists while T3 is better aligned with policy momentum. Assessors were evenly split between these two trajectories.
T2 vs T3	T2	T2	T2	T2	Both T2 and T3 aligned with policy momentum and landscape pressures but T2 was deemed to have broader coalition support.

Table 6

Comparison of model simulation results for the three policy trajectories.

Trajectory	Fuel-price change relative to T1	Demand multiplier	Projected rail growth	Change from T1, pp	Share of gap
T1: Flat tax	0.0%	1.00	31.0%	0.00	0.0%
T2: Moderate increase	5.0%	1.02	33.3%	2.26	8.1%
T3: Green-tax-shift	14.2%	1.05	37.2%	6.24	22.3%

political feasibility becomes entangled with technological deployment, infrastructure, behaviour, and economic change. Scenario practitioners often encounter the problem at a more specific point: when politically contingent tax, subsidy, or regulation assumptions must be expressed as model parameters. Changing the unit of assessment from the pathway to the policy trajectory builds on the comparative and causal reasoning of the feasibility literature while making it practical in scenario design. This means that the framework addresses the manageability of the inside view by narrowing the unit of assessment and the transparency problem by forcing the evidence and reasoning behind contested assumptions into a documented and comparable format. It does not, however, resolve the validation problem: feasibility judgement continues to rest on qualitative assessment contingent on evidence selection and analyst interpretation. Further, it does not translate feasibility into likelihood or probability estimates or fully address how actor interests and coalitions reorganise and evolve throughout the modelled period. Over longer periods, the political economy of a policy and actor interests may shift [1], making historically grounded comparison less secure. The framework should therefore be understood as a structured basis for selecting and documenting politically grounded scenario assumptions within a defined decision window, rather than as a prediction of the policy path that is most likely to occur.

Although the forecast adjustment is modest, a difference of two to six percentage points in projected rail demand growth can have meaningful impacts on downstream calculations. Fuel prices affect modal shifts through multiple channels in a transport model such as Samplan, most of which are not represented here, so the effects of the policy assumption propagate across transport modes. In the Samplan model, fuel costs affect the generalised cost of car travel, traffic volumes, and use of road networks in addition to rail demand. These outputs subsequently enter infrastructure appraisal through cost-benefit analysis, as well as downstream energy system modelling and emissions projections. Once a contested trajectory is quantified in a baseline, it is carried into each of these contexts, which is why the screening is useful before quantification.

The usefulness of the framework depends on the evidence available

for specifying and comparing trajectories. It is most readily applied where decision windows can be identified, policy processes are reasonably transparent, and actor positions can be mapped and reconstructed. Institutional differences can also affect which dimensions are most informative. For example, coalition bargaining may be especially important in coalition-based and multi-party political systems, including those common in the Nordic countries, whereas executive control, veto power, or party control may weigh more heavily elsewhere. Adjustment burden might similarly be more relevant in political systems that readily respond to voter dissatisfaction. More narrowly, some policies, such as subsidies, might not impose immediate and visible costs at all, which shows how some dimensions can be less relevant in particular settings.

There are additional limitations specifically tied to the case study. The Swedish rail case is a single historical demonstration, and the simplified demand model is only intended to illustrate how the framework can be used to parameterise assumptions for scenario design in a setting where seemingly neutral assumptions remain contested.

7. Conclusions

Policy assumptions have a substantial impact on model outputs, but are rarely assessed for political feasibility before quantification. Existing feasibility approaches are conceptually valuable but difficult to operationalise in conventional scenario practice. The framework developed here operationalises feasibility by narrowing the object of assessment to contested policy trajectories and connects historical and comparative evidence with an inside view of the actors, institutions, policy momentum, and landscape pressures interacting with the policy.

This translates feasibility analysis into a workflow for specifying consequential model inputs. It helps scenario teams and modellers define alternatives based on institutional theory and political economy, record the basis of comparison, and identify diverging interpretations. Its contribution lies in making the political judgement inherent in scenario design explicit, transparent, and structured.

The Swedish rail case demonstrates the workflow on a forecasting miss that has been attributed to policy assumptions made in the scenario design, specifically the excise tax on road fuels. The flat real road fuel tax path used in the Samplan baseline appeared less neutral once placed in its late-1990s political context. The framework supported alternative trajectories grounded in policy momentum, actor dynamics, institutional setting, and broader political and cultural trends. Their inclusion changed projected rail demand and showed how a single politically contingent assumption can affect model results.

CRediT authorship contribution statement

David Fornborg: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used OpenAI Codex for proofreading and improving readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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